

ISSUES IN THE MODELLING AND CONTROL OF FED BATCH FERMENTATION PROCESSES

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Abstract. In this paper alternative approaches to modelling and state estimation of a typical, industrial fermentation process will be described and comparative results of the application of the Kalman filter estimator to provide on-line estimation of the unmeasurable state variables will be given. Different techniques to control the product concentration of the process around a desired trajectory will be described. The several results are tabulated and compared, using a least squares criterion. A self organising system based on fuzzy logic control and model reference variable structure control will be illustrated as a possible candidate strategy to control the process.

Key Words. Biotechnology; fuzzy control; identification; Kalman filter; modelling; model reference control

1. INTRODUCTION

In controlling a fermentation process the most difficult problem is to determine the current state. This problem can be overcome if a suitable process model can be found for then an estimator can be used to determine the non-measurable variables (Jalel *et al* 1992). In this paper, an autoregressive (AR) technique is first adopted to develop a model using process data from the full scale production plant. The identification technique is a linear time invariant approach while fermentation is a highly non-linear time varying process. Therefore different techniques are investigated to create a general time varying model to represent the process over the whole operating range. Sequential modelling, time varying techniques and nonlinear difference equation models will be described, illustrated and compared for their suitability. A multi-layer neural network has been developed to model the state variables of the process. Inputs for the network are the on-line measurements while the outputs are the off-line measurements whose values are to be estimated. In the fermentation process one approach to control is to define a desired trajectory for the process and then to drive the process along the trajectory by controlling the amounts of nutrient feed. Two approaches have been investigated: self organising fuzzy logic and model reference variable structure control. The two approaches are described and the feasibility of their use in the fermentation process briefly illustrated.

2. THE AR MODEL

Initially, the Autoregressive (AR), identification approach, is adopted to develop a model for the fed batch process from the available data. Two AR models have been derived to describe the fermentation process. In the first, a model has been derived to express the relation between the inputs and the unmeasurable state variables, such as potency, residual carbon etc. The second model is used to express the relation between the state variables and the measured output oxygen uptake rate (RO_2).

2.1 Kalman Filter Estimator

The Kalman filter is used here to provide estimates of inaccessible variables that represent the biological and chemical state of the process. Here a model represented in the form of A, B, and C matrices and a measurement in the form of the model output oxygen uptake rate (RO_2) have been used to derive the Kalman filter.

3. SEQUENTIAL MODELLING

Since fermentation is a time varying process, a sequential modelling approach has been investigated in which the fermentation process is divided into

phases according to the operation time of the process. From manual inspection of the data and from a physical understanding of the process together with some experimentation it has been decided to divide the process into three phases. Each phase has been treated separately and a model for each has been derived. The unmeasurable state variables of the whole process have been taken to be the sequential combination of the three models. Of course, an important point is to find and implement the change-over between the three phases. Here, the change-over point is chosen by manual inspection of the data.

4. TIME-VARYING MODEL

In this section, different techniques are investigated to create a general time-varying model to represent the process over the whole operating range. This approach is based on generating a number of linear models to span the batch using historical input and output data. Since the coefficients of the model are updated along the batch, the time varying model is able to capture the process characteristics. The usefulness of such a model, able to span accurately a whole batch from start to finish, a range of batches and, after adaptation of numerical coefficients, a range of different fermentations, is obvious.

4.1 Polynomial Fitting Approach

This approach is based on the sequential modelling technique and is aimed to transfer the three linear models derived using the AR technique for the three phases into one time variant model which represents the whole process. The constants of the A and B matrices for the three models derived using the sequential modelling have been expressed in a time series polynomial. So instead of having three sets of constants for the coefficients of the A and B matrices a time variant polynomial has been derived which represent the variation constants as a function of time (more strictly, the age of the batch). The derived polynomial is used to represent and cover the time variation of the process over the whole operation.

4.2 Varying-Width Window

This approach is based on creating an overlapping AR models. The AR model is derived over a data window moved by a fixed step along the time axis of the process. For each of the (overlapping) windows a new AR model has been derived. At each moment of time, several overlapping AR models are available and the average of them is used to represent the process.

4.3 Generating a Time Invariant Nonlinear Ordinary Differential Equations (TINLODE)

This technique is based on generating a general nonlinear time varying model using the sequential linear models described previously as a starting point. As an illustration of the procedure, consider a second order system described as

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \quad (1)$$

If this linear model had been derived from a nonlinear system containing functions f_1, f_2 then the A and B constants used correspond to

$$\frac{\partial f_1}{\partial x_1} = a_{11}, \quad \frac{\partial f_1}{\partial x_2} = a_{12}, \quad \frac{\partial f_2}{\partial x_1} = a_{21}, \quad \frac{\partial f_2}{\partial x_2} = a_{22} \quad (2)$$

$$\frac{\partial f_1}{\partial u_1} = b_{11}, \quad \frac{\partial f_1}{\partial u_2} = b_{12}, \quad \frac{\partial f_2}{\partial u_1} = b_{21}, \quad \frac{\partial f_2}{\partial u_2} = b_{22} \quad (3)$$

With a simple nonlinear function defined by

$$\dot{x}_1 = q_1 x_1 + q_2 x_2 + q_3 x_1^2 + q_4 x_2^2 + q_5 x_1 x_2$$

is chosen.

Then from (2) and (3), we find that

$$a_{11}^{(1)} = q_1 + 2q_3 x_1^{(1)} + q_5 x_2^{(1)} \quad (4)$$

$$a_{11}^{(2)} = q_1 + 2q_3 x_1^{(2)} + q_5 x_2^{(2)} \quad (5)$$

$$a_{11}^{(3)} = q_1 + 2q_3 x_1^{(3)} + q_5 x_2^{(3)} \quad (6)$$

Where (4), (5) and (6) relate to phase 1, 2 and 3 respectively. In these equations all the variables are defined from the AR models and the off-line measurements, thus q_1, q_3 and q_5 can be estimated using a least squares estimation algorithm. The same procedure can be applied to estimate the parameters of the second function representing $\dot{x}_2$. The functions for $\dot{x}_1$ and $\dot{x}_2$ are used to represent the 2nd order nonlinear model of the process.

5. DIFFERENCE EQUATION MODEL

The concept, (Billings, 1980), is based on the nonlinear autoregressive moving average model given as

$$y(t) = F^* [y(t-1), y(t-2), \dots, y(t-n_y), u(t-d), \dots, u(t-d-n_u+1)]$$

where $F^*[\cdot]$ is some nonlinear function of u and y , n_u is the order of lagged inputs, n_y the order of lagged

outputs, d the time delay and l the degree of nonlinearity. Expanding $F^l[\cdot]$ as a polynomial and defining $V_1 = y(t-1), \dots, V_{ny} = y(t-n_y), V_{ny+1} = u(t-d), V_s = u(t-d-n_u+1)$ where $s=n_u+n_y$, the above equation can be expressed as

$$y(t) = \sum_{i=1}^l \theta_i V_i + \sum_{i=1}^s \sum_{j=1}^l \theta_{ij} V_i V_j + \dots + \sum_{i=1}^s \dots \sum_{m=1}^s \sum_{n=1}^l \theta_{i \dots mn} V_i V_j \dots V_m V_n$$

For the fed batch fermentation process, the parameters of the model are stored as a time variant set and used in connection with the Kalman filter to estimate the state variables. In a fermentation process there is a considerable amount of deviation between different batches. The algorithm has been modified and adopted to generate one model for more than one batch of data. The model will hopefully represent the different batches over the whole operating range of the process. In this contribution, the algorithm has been applied and tested on twenty different industrial batches in which one time varying model has been derived to represent all the batches.

5.1 The Use Of Off-line Measurements In State Estimation

During a fermentation batch, there is a regular sampling at a fixed period of time of the substrate inside the fermenter. This is used in the laboratory for analyzing and calculating the state variables of the process. The off-line measurements obtained through the laboratory are usually some hours lagging the actual value of the states inside the fermenter. In this contribution the off-line measurements have been used to update and improve the model output derived using the difference equation technique. The approach is based on continuously using the available off-line measurement which is few hours behind the current situation to regressively calculate and update model output at the current period of time. The adaptive time varying model that has been derived using this technique has improved accuracy compared with model using only on line data.

6. NEURAL NETWORK

In this contribution, the neural network approach has been adopted to estimate the state variables in a fed batch fermentation process. A neural network was adopted for the on-line estimation of the product concentration in the fed batch fermentation process with the neural network taking on the task of both modelling and state estimation. In this work a two layered, fully connected, feed-forward networks learning the fermentation process by standard back-propagation algorithm has been used to model the process (Rumelhart *et al*, 1986).

7. USING A FUZZY LOGIC CONTROLLER IN THE FED BATCH FERMENTATION PROCESS

The basic design of a self organizing fuzzy logic controller (SOFLC) has been described in several papers (Procky and Mamdani, 1979). It consists of two levels, the first containing the simple fuzzy logic controller and the second containing the self organizing mechanism, Fig.1. SOFLC has been applied to the simulated control of the trajectory of the fed batch product, by controlling the amount of carbon fed to the fermenter and using the derived AR model, described previously, to estimate the process state. Fig. 2 shows the desired trajectory and the product concentration of the process. From this figure it is possible to see that SOFLC has achieved good control in following the desired trajectory.

8. MODEL REFERENCE VARIABLE STRUCTURE CONTROL (MRVSC) IN FERMENTATION TECHNOLOGY

There have been major studies in the use of variable structure control (VSC) in model-following and model reference adaptive control where most applications are in the Aerospace and Robotics field (Young, 1978). In this work the feasibility of using MRVSC to control the product concentration of the fed batch fermentation process has been investigated. The plant is simulated using an AR model while in reality the state of the plant can be estimated using the Kalman filter. The desired model is derived using the AR approach with five inputs representing the reference inputs to the process. The resulting responses of the plant and the model are illustrated in Fig. 3. From this figure it is possible to see that a good result has been achieved and this technique is a good candidate to control the product of the fed batch process.

9. CONCLUSION

In this paper different approaches for modelling the fed batch fermentation process have been investigated. Reasonable results have been achieved using the sequential modelling approach. However the most difficult aspect is finding the change over point between the phases. Generating a time varying model from the linear AR model is a very attractive approach, since the time varying model is general and can cope with different situations. Good results have been achieved through the polynomial fitting, varying-width window and TINLODE approaches. A nonlinear difference equation approach has been found to provide impressive modelling and estimation of the state variables due to its ability to

generate a general model of the different batches of the process. The modelling results of the different techniques are tabulated and represented in table 1. Controlling the states of the process around a defined trajectory is achieved using two techniques, SOFLC and MRVSC. The two approaches have achieved good results due to their ability to handle the variations and disturbances in the model parameters. The feasibility of using these approaches in fermentation technology is currently being evaluated. The excellent performance of the neural network in this task, has resulted in the neural network model being chosen for implementation in the system intended for future supervisory modelling and control.

10. REFERENCES

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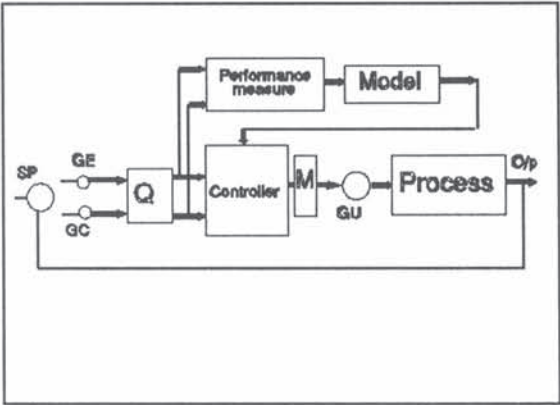


Fig. 1. SOFLC

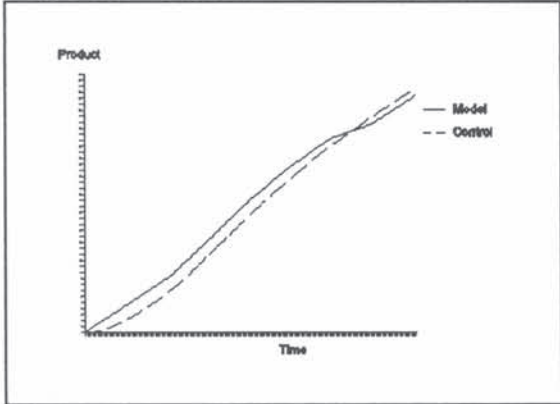


Fig. 2. Product -- SOFLC

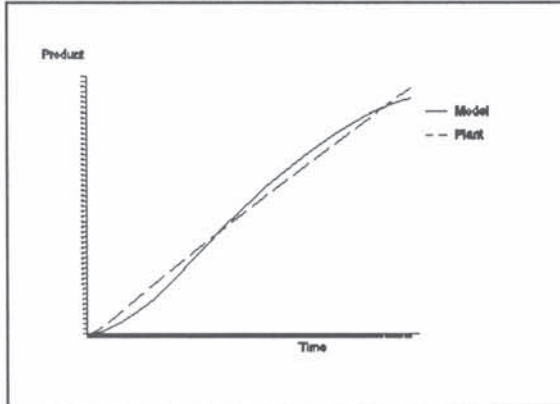


Fig. 3. Product -- MRVSC

Table 1 Performance of the Models

Model Type	Mean error between fit and off-line measurement for product
Simple AR	2.3528, 2.268 (Kalman Filter)
Sequential modelling	1.1821, 0.0394 (Kalman Filter)
Time varying 1. Polynomial, 2. Window, 3. TINLODE	1. (1.0572), 2. (1.1276), 3. (0.2251)
Difference equations models	0.4668
Neural network	0.0099